Lakshman Shanmugam

+91 6380083611 | slakshman2004@gmail.com | LinkedIn | GitHub

Professional Summary

AI/ML Engineer experienced in developing LLM-powered applications, agentic workflows, RAG systems, and machine learning solutions across cybersecurity, GRC, and healthcare. Proficient in Python, PyTorch, LangChain, LangGraph, NLP, and deep learning, with hands-on experience in API integration, intelligent automation, and AI application development.

Technical Skills

Languages: Python, Java, R, SQL (MySQL, PostgreSQL), Dart
Machine Learning: TensorFlow, PyTorch, Keras, NLTK, NumPy, Matplotlib, Scikit-learn, LLMs
Cloud & Data Tools: Git, Docker, AWS, MongoDB, MS Excel
Others: Algorithms, Flutter, Power BI, Langchain, Jupyter Notebooks

Experience

Junior Engineer - AI ML Nov. 2025 – Present
SQ1 Security Technology Pvt Ltd *Chennai, TN*

- Developed AI/ML solutions across cybersecurity, Governance, Risk, and Compliance (GRC), and healthcare products, integrating machine learning models and LLM-powered workflows into existing applications.
- Built a LangGraph-based multi-agent proof of concept for Stakflo to automate evidence collection, normalization, validation, and control mapping using API-integrated tools and structured evidence schemas.
- Contributed to Scani5 vulnerability management through Common Platform Enumeration (CPE) generation and vulnerability visibility workflows.
- Worked on Diagnox medical NLP and clinical coding pipelines, developing structured information extraction and LLM-based classification workflows and evaluating prompt and model performance.
- Developed Zycato purpose-mapping and retrieval workflows, integrating LLM-based decision logic, PostgreSQL, and vector-based retrieval to support purpose matching and unmatched-purpose handling.

Machine Learning Engineer Intern May. 2025 – Aug. 2025
Vcodez *Chennai, TN*

- Trained and Tested regression and classification models with public datasets from Kaggle and Hugging Face.
- Performed Exploratory Data Analysis on public datasets.
- Developed an AI Powered Legal Document Intelligence System with 4 features backed by a Finetuned BERT and Logistic Regression for NLP tasks.

Generative AI Developer Intern May. 2024 – Sep. 2024
Trueinfo Labs *Chennai, TN*

- Collaborated with stakeholders to design and perform AI-driven content analysis with 97% accuracy by exploiting FFmpeg, Whisper, and GPT-4, visualized through a Streamlit interface
- Spearheaded initiatives to accelerate data processing by 25% with AI-driven OCR and Web scraping pipelines
- Teamed up to refine data retrieval speed by 20% leveraging Selenium, BeautifulSoup, and MongoDB workflows

Projects & Publications

TAD-LSTM for Financial Time-Series Forecasting - [View Paper]

- Published in Neural Computing and Applications, Springer Nature, Volume 38, Article 178, 2026.
- Developed and implemented a novel Temporally Adaptive Dropout LSTM (TAD-LSTM) framework with six variants for enhanced financial time-series forecasting, using adaptive regularization to improve stock price prediction and generalization across NIFTY 50, NSE stocks, and AAPL datasets, evaluated using MAE, RMSE, MAPE, and R^2 metrics across multiple prediction horizons.

Optimized Multi-Feature Gated MLP for Lung Disease Detection - [View Paper]

- Accepted in IEEE Access Journal, Volume 13, 2025. Published in IEEE Xplore
- Developed a novel lung disease detection model using a Gated MLP framework integrated with Grey Wolf Optimization (GWO) and multi-feature extraction (ResNet50, HOG, GLCM, MRWT, ViT, SP-GNN), enhanced with cumulative learning and recursive optimization, achieving 98% accuracy.

Education

Vellore Institute of Technology Chennai, TN

B. Tech in Electronics and Computer Engineering | CGPA: 8.5/10.0 *2021 – 2025*